TerraBit: Low-Bit Quantization of Geospatial Embedding Lakes 🌊

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Abstract

Geospatial foundation models produce planet-scale *embedding lakes*—large, centralized stores of precomputed vectors analogous to data lakes—whose storage and retrieval costs dominate deployment budgets. Recent systems distribute global embedding fields at `int8` precision, but the quality–cost trade-off below 8 bits remains unexplored for remote sensing embeddings. We present **TerraBit**, a post-hoc quantization study spanning `float16` through 1-bit `binary` on 49,785,116 real 1024D embeddings (*183 GiB*), evaluated with label-free neighborhood-preservation metrics grounded in intrinsic-dimension analysis. `Int4` retains 91% kNN recall at $6.3\times$ compression; `binary` achieves $16.5\times$ compression with 65% recall, sufficient for coarse-stage candidate generation.

1 Introduction

Geospatial foundation models [Mendieta et al., 2023, Jakubik et al., 2024, Clay Foundation, 2025] produce high-dimensional embeddings for retrieval, change detection, deduplication, and spatial indexing, forming *embedding lakes*—large, centralized vector stores analogous to data lakes. At planet scale, storage and memory bandwidth—not model accuracy—become the deployment bottleneck. A corpus of 50M embeddings at d=1024 in `float32` occupies *183 GiB*; halving per-vector cost saves nearly *100 GiB* with no retraining.

Two recent systems highlight this tension. AlphaEarth releases global embedding fields cast to `int8` via post-hoc calibration [Brown et al., 2025]. TESSERA applies quantization-aware training (QAT) to a learned 128D bottleneck and distributes the result at `int8` [Feng et al., 2025]. Both treat `int8` as the practical floor. **TerraBit** asks: *how far below `int8` can we push post-hoc quantization while preserving nearest-neighbor structure?*

Figure 1 previews the answer. We sweep seven bit-widths from `float16` to 1-bit `binary` on a real 49.8M-embedding corpus and measure neighborhood preservation with label-free metrics. The frontier is smooth: `int4` retains 91% kNN recall at $6.3\times$ compression, while `binary` reaches $16.5\times$ with recall sufficient for shortlist retrieval [Wang et al., 2018].

Our contributions are threefold: (1) an intrinsic-dimension-guided evaluation framework combining ID diagnostics [Levina and Bickel, 2004, Facco et al., 2017] with label-free retrieval fidelity metrics; (2) a systematic post-hoc quantization study (`float16`, `fp8`, `int8`, `int4`, `int3`, `int2`, `binary`) on 49.8M real geospatial embeddings; and (3) a schema-preserving parallel Parquet transformation pipeline that processes the full corpus in 45 min at 18,359 rows/s.

2 Related Work

Geospatial foundation models. Recent work trains large vision encoders on satellite imagery and releases pretrained embeddings as data products [Mendieta et al., 2023, Jakubik et al., 2024, Clay

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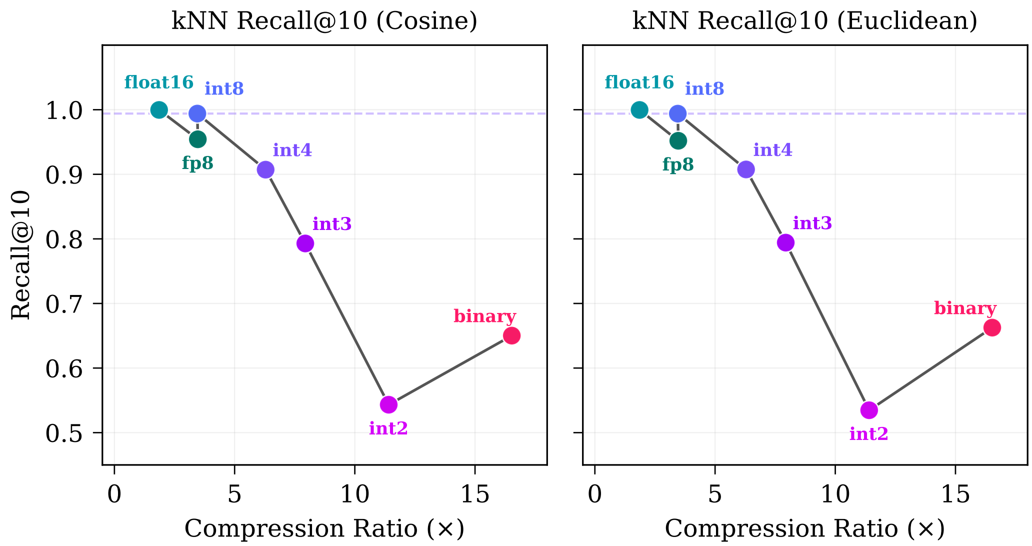


Figure 1: **Quality–compression Pareto frontier.** kNN recall@10 (cosine and Euclidean) vs. compression ratio on 49,785,116 geospatial embeddings. `Int8` (dashed line) is the current practical baseline [Brown et al., 2025, Feng et al., 2025]. `Int4` preserves 91% recall at $6.3\times$; `binary` reaches $16.5\times$ with useful shortlist-level recall.

Foundation, 2025]. AlphaEarth [Brown et al., 2025] and TESSERA [Feng et al., 2025] both distribute global `int8` embedding fields, recognizing that storage cost matters as much as model quality at deployment scale.

Vector quantization and retrieval. Product quantization [Jégou et al., 2011], learned anisotropic quantization [Guo et al., 2020], and locality-sensitive hashing [Indyk and Motwani, 1998] compress embeddings for approximate nearest-neighbor search. FAISS [Johnson et al., 2021] implements these at billion scale. LLM.`int8`() [Dettmers et al., 2022] demonstrates that aggressive quantization preserves model quality in language; Matryoshka representations [Kusupati et al., 2022] show that embedding dimensions can be truncated with graceful degradation.

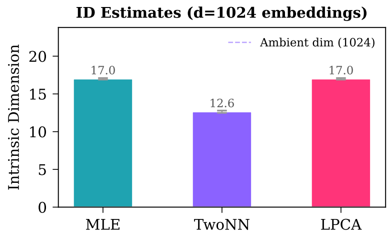
Positioning. Prior vector-compression work targets web-scale text/image embeddings or model-weight quantization. **TerraBit** focuses on the less-studied regime of post-hoc quantization of geospatial embedding lakes, where spatial metadata must be preserved alongside vectors, and label-free evaluation is the practical reality.

3 Method

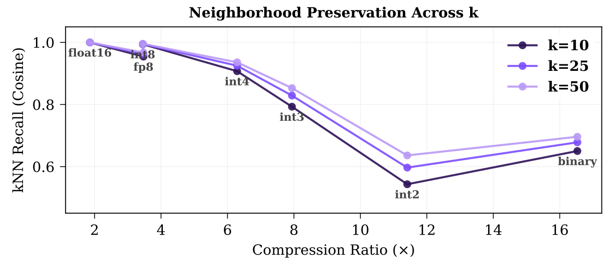
Problem setup. Given a corpus $X = \{x_i\}_{i=1}^N$, $x_i \in \mathbb{R}^d$, stored in partitioned Parquet files with spatial/temporal metadata columns, we apply a quantizer $\hat{x}_i = \mathcal{Q}_m(x_i)$ under method m and evaluate each method by storage footprint, neighborhood preservation, and end-to-end throughput. Our corpus comprises 49,785,116 embeddings (d=1024) produced by the Clay v1.5 foundation model [Clay Foundation, 2025], a ViT trained via masked autoencoding on multi-sensor satellite imagery.

Intrinsic-dimension diagnostics. Before compression, we estimate the intrinsic dimension (ID) of the embedding manifold using MLE [Levina and Bickel, 2004], TwoNN [Facco et al., 2017], and local PCA participation ratio across preprocessing variants (raw, centered, L2-normalized). ID estimates guide expectations for viable compression depth: if the data manifold has effective dimension $\hat{d} \ll d$, aggressive quantization should not collapse local neighborhood structure.

Quantization methods. We evaluate seven post-hoc methods spanning 16 bits to 1 bit: `float16` (IEEE half-precision truncation); `fp8` E4M3 (8-bit floating point with 4-bit exponent); int$b$ for



(a) ID estimates. All estimators converge to $\hat{d} \approx$ 13–17, far below d=1024.



(b) kNN recall (cosine) across $k \in \{10, 25, 50\}$. Recall degrades gracefully as bit-width decreases.

Figure 2: **Intrinsic dimension and neighborhood preservation.** (a) The embedding manifold has $\hat{d} \approx$ 13–17, motivating aggressive quantization of redundant dimensions. (b) Neighborhood preservation declines smoothly from `int8` to `binary`, enabling application-specific operating-point selection.

$b \in \{8, 4, 3, 2\}$ (per-dimension affine quantization with stored scale/zero-point vectors, followed by fixed-width bit packing for $b < 8$); and `binary` (sign-bit encoding, $\mathrm{sign}(x_i) \in \{0, 1\}^d$, packed to $d/8$ bytes, supporting Hamming-distance search [Wang et al., 2018]).

Evaluation metrics. We report label-free, retrieval-oriented metrics: **kNN recall@**k ($k \in \{10, 25, 50\}$) under cosine and Euclidean distance, measuring the fraction of true k-nearest neighbors preserved after quantization; **cosine correlation**, the Spearman rank correlation of pairwise cosine similarities (original vs. quantized); and **reconstruction cosine similarity** and MSE (when dequantization is available).

Transformation pipeline. The pipeline processes Hive-partitioned Parquet files in parallel. For each file, it reads the embedding column, applies quantizers, replaces only that column, and writes output preserving directory structure, non-embedding columns, and schema. Quantization parameters (scale, zero-point, method, bit-width) are stored in Parquet footer metadata for downstream decompression.

4 Experiments

Dataset. We evaluate on 49,785,116 embeddings (d=1024, `float32`) from a geospatial foundation model, stored across 1,021 Parquet files totaling *183 GiB*. Files are Hive-partitioned by level-2 geohash (511 cells), year, and month. Each record carries spatial metadata (geohash, bounding box, WKB geometry), temporal metadata, and provenance identifiers alongside the embedding vector.

Protocol. We evaluate along two axes: quality and systems efficiency. For quality, we compute label-free fidelity metrics on a 20,000-row reservoir sample (reconstruction metrics are exact over all 49,785,116 rows). For systems, we run the full-corpus transformation over all 7 methods with 8 parallel workers, reporting wall-clock time and I/O throughput.

ID estimation on a 10,000-sample subset yields MLE $\hat{d}$ = 17.0, TwoNN $\hat{d}$ = 12.6, and LPCA $\hat{d}$ = 17.0 (Figure 2a), confirming substantial structural redundancy relative to ambient dimension d=1024.

Quality–compression frontier. Table 1 summarizes quality across all methods. `Int8` yields near-lossless retrieval (recall@10 = 0.99) at 3.5× compression, matching the precision used by AlphaEarth and TESSERA [Brown et al., 2025, Feng et al., 2025]. `Int4` provides a strong balanced operating point: 6.3× compression with 91% kNN recall. `Binary` achieves the highest compression (16.5×) with recall@10 of 0.65—suitable as a coarse first-pass filter before re-ranking with higher-fidelity representations [Wang et al., 2018, Jégou et al., 2011].

Table 1: **Quality–compression results on** $49{,}785{,}116$ **embeddings.** Reconstruction cosine is exact over all rows; kNN recall and cosine correlation are sampled from a 20,000-row reservoir. Best compression in **bold**.

Method	Ratio (×)	Recon. Cos ↑	C@10 ↑	E@10 ↑	Cos Corr ↑
`float16`	1.86	1.00	1.00	1.00	1.00
`fp8`	3.47	1.00	0.95	0.95	1.00
`int8`	3.45	1.00	0.99	0.99	1.00
`int4`	6.29	1.00	0.91	0.91	1.00
`int3`	7.94	0.98	0.79	0.79	0.99
`int2`	11.41	0.91	0.54	0.53	0.95
`binary`	**16.53**	0.49	0.65	0.66	0.79

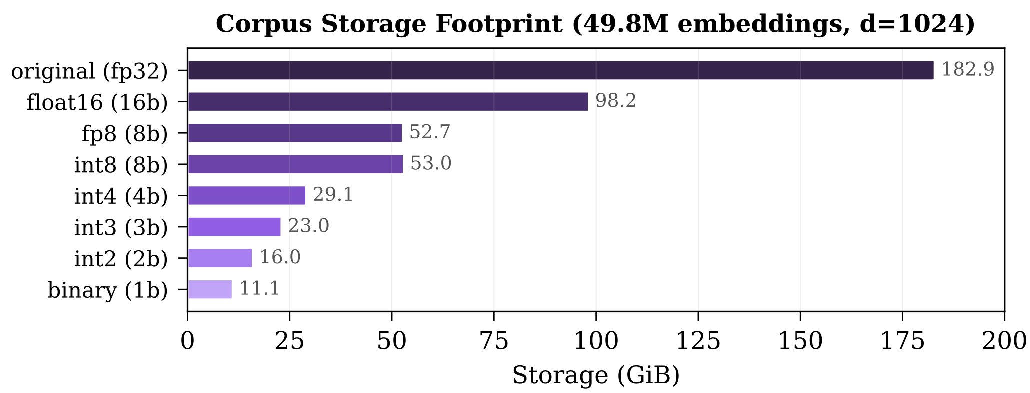


Figure 3: **Storage footprint per quantization method.** Original `float32` corpus occupies *183 GiB*. `Binary` reduces this to *11 GiB* (16.5×); `int4` to *29 GiB* (6.3×).

Storage and throughput. Figure 3 shows the per-method storage footprint. `Binary` reduces the *183 GiB* corpus to *11 GiB*; `int4` to *29 GiB*. The full pipeline (all 7 methods, 8 workers) processes the entire corpus in $45.2\,\mathrm{min}$ at 18,359 rows/s and $107\,\mathrm{MiB/s}$ output throughput.

Discussion. Three findings emerge. First, ID estimates of 13–20 (vs. ambient $d{=}1024$) explain why aggressive quantization does not immediately collapse neighborhood structure: most per-dimension variance is redundant for local geometry. Second, the quality decline from `int8` to `int2` is gradual (Figure 2b), enabling practitioners to select operating points based on their recall requirements versus storage budget. Third, `binary` quantization is not quality-preserving in a strict sense, but its $16.5\times$ compression and compatibility with Hamming-distance search make it attractive for first-pass candidate generation in multi-stage retrieval pipelines [Wang et al., 2018].

Limitations. We use label-free neighborhood-preservation proxies rather than task-specific metrics (e.g., land-cover classification accuracy). kNN recall and cosine correlation are computed on a reservoir sample, not the full corpus. We also do not benchmark end-to-end ANN index build/query latency [Johnson et al., 2021]. Future work should add labeled downstream evaluation, confidence intervals across seeds, and hardware-specific retrieval benchmarks.

5 Conclusion

TerraBit provides a systematic compression study for geospatial embedding lakes. On 49.8M real embeddings, low-bit quantization yields a smooth quality–cost frontier: `int4`/`int8` are strong balanced choices; `binary` is viable for coarse retrieval stages.

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